Computer System Security
LINFO2347

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So, what is this course about?

- Introduction to the security of computer systems
- Topics covered (not in that order):
  - What is a secure system?
  - Software security: buffer overflow, code injection
  - Web security: SQL injection, cross site scripting
  - Network security:
    - Attacks: cache poisoning, DoS, Botnets,…
    - Defenses: firewalls, NATs
  - Intrusion detection systems,…
  - Spam & phishing
  - Secure communication: TLS, certificates, DNS over TLS/HTTPS

Prerequisites

- Knowledge of computer architectures and networks required
  - IP network protocols: ARP, IP, TCP/UDP, DNS, HTTP
  - Computer architecture: how does a program look like in machine language, what is virtual memory
  - Basic knowledge of C

- If you don't remember anymore how a TCP packet looks like or how strings work in C, it is strongly recommended that you take a look at the respective courses (LINFO1341 or LINFO2147, LINFO1252 or LINFO2241,...)

Material

- On Moodle
- We will have
  - Slides
  - Exercises
  - Instructions for projects
  - Links to relevant publications, websites, etc. that you have to read

Assessment

- There will be an exam on the topics addressed in the lectures, counting for 50% of the final grade
- Several project activities: 50% of the final grade
  - Write code and report
  - Some of them in groups, some individually
- Exercises will not be assessed, but they will prepare you for the projects and the exam

- The project activities cannot be done or repeated for the August session! You will keep the grade that you received in the June session.

- See https://uclouvain.be/en-cours-2025-linfo2347

Disclaimer

- Some of the attacks that we will see in this course are very easy to execute
- Even a “harmless” attack (e.g., a port scan) can cause damage, for example by crashing the target computer or by disturbing the network
  - Always ask the target of the attack for permission
  - Forbidden in most networks (UCLouvain, your ISP,...)